Chapter 8

The Calculus of Variations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 8; printed p. 427; PDF page 444.

Chinese reference: 中译本对应章：变分法，书内第 303 页起。

Study edition / 英文重构学习稿。 This chapter follows the source's section order. Parenthetical theorem identifiers refer to Evans's section-local numbering; the displayed numbers belong to these notes. Exposition, proof reconstructions, Chinese annotations, solutions and TikZ drawings are editorial contributions, not verbatim text. The chapter has not received independent mathematical review.

The central question is how an energy produces a PDE. First variations give the equation; coercivity and lower semicontinuity produce minimizers; constraints replace equations by multipliers or variational inequalities. Minimax methods locate further critical points, while invariance yields conservation and monotonicity laws. Unless stated otherwise, U is bounded with smooth boundary and $1 < q < \infty$. A prescribed trace is represented by a fixed Sobolev lift, and all existence claims below require a finite-energy competitor. External regularity and divergence-solving theorems are identified where used.

8.1 Introduction

8.1.1 First variation and the Euler–Lagrange equation

For a smooth Lagrangian $L(p, z, x)$ define

$$I[u] = \int_U L(Du, u, x)\, dx, \qquad u - g \in W_0^{1,q}(U). \tag{8.1}$$

Here $g \in W^{1,q}(U)$ is a lift of the boundary data. An energy is a functional on functions; its derivative is initially a linear functional on variations, not automatically a pointwise differential operator. For a smooth minimizer u and $\phi \in C_c^\infty(U)$, the curve $i(s) = I[u+s\phi]$ has $i'(0) = 0$. Differentiation and integration by parts give

$$\delta I[u](\phi) = \int_U L_p(Du, u, x) \cdot D\phi + L_z(Du, u, x)\phi = 0, \tag{8.2}$$

so the Euler–Lagrange equation is

$$-\operatorname{div} L_p(Du,u,x) + L_z(Du,u,x) = 0. \tag{8.3}$$

For smooth functions and compactly supported variations, the derivatives are bounded on the compact ranges involved; this justifies differentiating inside the integral. The Sobolev justification appears in §8.2.

Expanding the equation shows its quasilinear principal part:

$$-L_{p_ip_j}u_{ij} - L_{p_iz}u_i - L_{p_ix_i} + L_z = 0.$$

Thus convexity of L in the gradient variable determines ellipticity. The standard examples, in the source's order, are

$L(p,z,x)$	Euler–Lagrange equation
$\frac{1}{2}\lvert p\rvert^2$	$-\Delta u = 0$
$\frac{1}{2}a^{ij}(x)p_ip_j - f(x)z$	$-\partial_i(a^{ij}u_j) = f$
$\frac{1}{2}\lvert p\rvert^2 - F(z),\ F' = f$	$-\Delta u = f(u)$
$(1+\lvert p\rvert^2)^{1/2}$	$\operatorname{div}\dfrac{Du}{\sqrt{1+\lvert Du\rvert^2}} = 0$

For the second example $A = A^T$; no arbitrary drift is generated by a quadratic unweighted Dirichlet energy. The last integrand measures graph area. Its Hessian has eigenvalues $(1+|p|^2)^{-1/2}$ transverse to p and $(1+|p|^2)^{-3/2}$ parallel to p: it is strictly convex, but not uniformly convex on all gradient space.

8.1.2 Second variation and the Legendre condition

For a smooth local minimizer, $i''(0) \geq 0$, hence

$$Q_u[\phi] = \int_U L_{p_ip_j}\phi_i\phi_j + 2L_{p_iz}\phi_i\phi + L_{zz}\phi^2\,dx \geq 0. \tag{8.4}$$

All coefficients are evaluated at (Du,u,x).

Theorem 8.1 (Necessary gradient convexity; source §8.1.3)
For every $x \in U$ and $\xi \in \mathbb{R}^n$, $L_{p_ip_j}(Du(x),u(x),x)\xi_i\xi_j \geq 0$.

Expanded proof (small amplitude, oscillatory gradient).

Let ρ be the periodic triangular function of period one, with $|\rho'| = 1$ a.e., and take $\phi_\varepsilon = \varepsilon\rho(x\cdot\xi/\varepsilon)\zeta(x)$, $\zeta \in C_c^\infty(U)$. Smooth approximation makes (8.4) valid for this compactly supported Lipschitz function. Its value is $O(\varepsilon)$ and $D\phi_\varepsilon = \rho'\xi\zeta + O(\varepsilon)$, uniformly on the fixed support except at the harmless corners. The mixed and zeroth-order terms vanish as $\varepsilon \to 0$, while $(\rho')^2 = 1$ a.e. gives $\int L_{p_ip_j}\xi_i\xi_j\zeta^2 \geq 0$. If this continuous

偏导数与全导数

L_{x_i} 只对显式的 x 变量求偏导；$\partial_i[L_{p_i}(Du,u,x)]$ 还包含 D^2u 与 Du 的链式项。Noether 公式中也必须保持这个区别。

极小曲面的退化

面积密度在大梯度方向的最小曲率约为 $|p|^{-3}$。严格凸不能代替一致椭圆；面积能量还只有线性增长，不属于后面的超线性强制性框架。

coefficient were negative at a point, choosing ζ in a small neighborhood would contradict the inequality. This is positivity along the actual gradient of u, not global convexity of L at all (p, z, x), and it is necessary rather than sufficient for a minimum. □

8.1.3 Systems, null Lagrangians, and the Piola identity

For $u : U \to \mathbb{R}^m$ write $P_{\alpha i} = \partial_i u^\alpha$. The first variation uses the Frobenius product $A : B = \sum_{\alpha,i} A_{\alpha i} B_{\alpha i}$:

$$\delta I[u](\phi) = \int_U L_P : D\phi + L_z \cdot \phi, \qquad -\partial_i L_{P_{\alpha i}} + L_{z_\alpha} = 0.$$

A null Lagrangian is one whose Euler–Lagrange expression vanishes for every smooth map. It need not have constant density or zero integral.

Theorem 8.2 (Boundary dependence; source §8.1.4, Theorem 1)
If L is null and $u, v \in C^2(\overline{U}; \mathbb{R}^m)$ have the same boundary values, then $I[u] = I[v]$.

Expanded proof (a homotopy with fixed trace).

Put $u_s = v + s(u - v)$ for $0 \le s \le 1$. Its energy derivative is $\int L_P(Du_s, u_s, x) : D(u - v) + L_z(Du_s, u_s, x) \cdot (u - v)$. Integration by parts has zero boundary term because $u - v = 0$ there. The remaining Euler–Lagrange expression vanishes by nullity, so $\frac{d}{ds} I[u_s] = 0$. Integrating in s proves the claim. □

Lemma 8.3 (Piola identity; source divergence-free rows lemma)
For smooth $u : U \to \mathbb{R}^n$, with the cofactor matrix in the same row-column convention as Du,

$$\partial_i (\operatorname{cof} Du)_{\alpha i} = 0. \tag{8.5}$$

Consequently $L(P) = \det P$ is a null Lagrangian (source Theorem 2).

Expanded proof (antisymmetry rather than invertibility).

The entry $(\operatorname{cof} Du)_{\alpha i}$ is

$$\frac{1}{(n-1)!} \epsilon_{\alpha \alpha_2 \cdots \alpha_n} \epsilon_{i i_2 \cdots i_n} u^{\alpha_2}_{i_2} \cdots u^{\alpha_n}_{i_n}.$$

Differentiate in i and sum. In each resulting term the second derivative is symmetric in i, i_j, whereas the spatial alternating symbol is antisymmetric in those indices; paired terms cancel. For $n = 1$ the cofactor is the constant 1. Since $\partial(\det P)/\partial P_{\alpha i} = (\operatorname{cof} P)_{\alpha i}$, (8.5) is precisely the Euler–Lagrange identity. No division by $\det Du$ occurs, so singular gradients are included. □

小振幅仍有大梯度

$\phi_\varepsilon = O(\varepsilon)$，但 $D\phi_\varepsilon = O(1)$。高频测试分离出二阶变分的主部；仅看函数值的接近不能忽略梯度振荡。

In two dimensions it reads $\operatorname{cof} Du = \begin{pmatrix} u^2_y & -u^2_x \\ -u^1_y & u^1_x \end{pmatrix}$; each row divergence is equality of mixed partial derivatives. All minors enjoy analogous cancellation, a fact used in the weak continuity argument below.

余子式的转置约定

这里 Du 的行是分量、列是空间方向；Piola 恒等式对每行取散度。矩阵恒等式为 $P(\operatorname{cof} P)^T = (\det P)I$，不能混用伴随矩阵与余子式矩阵。

8.1.4 An analytic route to Brouwer's fixed point theorem

Theorem 8.4 (Brouwer; source §8.1.4, Theorem 3)
Every continuous map $F : \overline{B} \to \overline{B}$, $B = B(0,1) \subset \mathbb{R}^n$, has a fixed point.

Expanded proof (no retraction, including the smoothing step).

A smooth retraction $r : \overline{B} \to \partial B$ would have the same boundary data as the identity. Nullity gives $\int_B \det Dr = |B|$. But $|r|^2 = 1$ implies $(Dr)^T r = 0$, so $\det Dr = 0$, a contradiction. If a continuous retraction existed, extend it by $r(x) = x$ outside B. It is nonzero everywhere and equals the identity outside a compact set. Convolution with an even radial mollifier gives smooth r_ε that remains nonzero for sufficiently small ε: uniform approximation on a fixed compact ball preserves the positive minimum of $|r|$, and farther away convolution reproduces x exactly. For $|x| \geq 2$, $r_\varepsilon(x) = x$ if $\varepsilon < 1$. Thus $2r_\varepsilon/|r_\varepsilon|$ is a smooth retraction from $\overline{B(0,2)}$ onto its boundary, the same contradiction after rescaling. Finally, if F had no fixed point, the ray from $F(x)$ through x would meet the sphere in a continuous point $r(x)$ with $r(x) = x$ on ∂B. Explicitly, for $a = F(x)$ and $d = x - a \neq 0$, take

$$r(x) = a + \lambda d, \qquad \lambda = \frac{-a \cdot d + \sqrt{(a \cdot d)^2 + |d|^2(1 - |a|^2)}}{|d|^2}.$$

The forward exit parameter is at least one; the formula is continuous because $d \neq 0$. It supplies the forbidden continuous retraction. □

8.2 Existence of Minimizers

8.2.1 Coercivity and the direct method

Fix $g \in W^{1,q}(U;\mathbb{R}^m)$ and the affine class $\mathcal{A} = g + W^{1,q}_0(U;\mathbb{R}^m)$, $1 < q < \infty$. The direct method separates three tasks: boundedness of minimizing sequences, closure of the admissible class under the chosen convergence, and a lower bound for the energy at the limit. Suppose

$$L(P,z,x) \geq \alpha|P|^q - \beta, \qquad \alpha > 0,\ \beta \geq 0. \tag{8.6}$$

A bounded energy sequence then has bounded Du_k in L^q. Poincaré applied to $u_k - g$ bounds u_k in $W^{1,q}$. Reflexivity gives a weakly convergent subsequence. The affine class is weakly closed: the continuous trace map has weakly closed kernel, or use that a norm-closed linear subspace is weakly closed by Hahn–Banach. The missing issue is weak lower semicontinuity:

$$u_k \rightharpoonup u \text{ in } W^{1,q} \quad \Longrightarrow \quad I[u] \le \liminf_k I[u_k].$$

Strong norm continuity alone does not imply this inequality.

三个条件各做什么

强制性给出有界极小化序列；反射性给出弱子列；弱闭性保住边界和约束；弱下半连续性才保证极限达到下确界。四步不能互相替代。

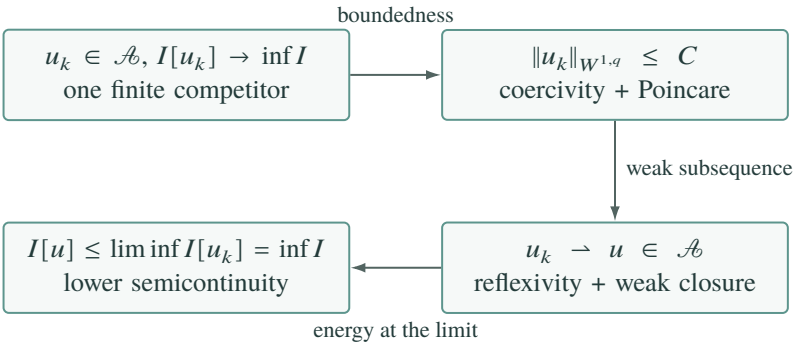


Diagram T8.1. The direct method keeps feasibility and energy in the limit. The arrows represent separate arguments. A finite-energy competitor makes the infimum meaningful; coercivity and Poincare give boundedness. Weak closure preserves constraints, while lower semicontinuity supplies the minimizing inequality.

8.2.2 Convexity in the gradient and weak lower semicontinuity

Theorem 8.5 (Lower semicontinuity; source §8.2.2, Theorem 1)
Let L be smooth on gradient and value space times $\overline{U}$, bounded below, and convex in P for each (z, x). Then I is sequentially weakly lower semicontinuous on $W^{1,q}(U;\mathbb{R}^m)$. No convexity in z is required. The scalar case is the source Theorem 1; the vector version is used in Theorem 5.

Expanded proof (linearizing only the weakly convergent variable).

Add a constant so $L \ge 0$. Pass to a subsequence realizing the liminf; if it is infinite there is nothing to prove. Compact embedding gives $u_k \to u$ strongly in L^q, and a further subsequence converges a.e. For $\varepsilon > 0$, Egoroff gives a set E_ε where this convergence is uniform, with $|U \setminus E_\varepsilon| < \varepsilon$. Intersect it with $\{|u| + |Du| \le \varepsilon^{-1}\}$ to obtain G_ε, whose complement has measure tending to zero. Convexity gives

$$\begin{aligned} I[u_k] \ge & \int_{G_\varepsilon} L(Du, u_k, x) \\ & + \int_{G_\varepsilon} L_P(Du, u_k, x) : (Du_k - Du). \end{aligned}$$

On this set all arguments except Du_k lie in a fixed compact range. The first integral converges uniformly in its integrand. In the second, $L_P(Du, u_k, x)$ converges uniformly to $L_P(Du, u, x)$. Its difference pairs with a uniformly bounded L^q sequence and tends to zero by Hölder; its fixed limit pairs to zero by weak convergence. Thus $\liminf I[u_k] \geq \int_{G_\varepsilon} L(Du, u, x)$. To exhaust U without assuming these sets are nested, fix $R < \infty$:

$$\int_{G_\varepsilon} L(Du, u, x) \geq \int_U \min\{L(Du, u, x), R\} - R|U \setminus G_\varepsilon|.$$

Let $\varepsilon \downarrow 0$, then $R \uparrow \infty$ by monotone convergence. This also covers an infinite limit energy. □

Theorem 8.6 (Existence and uniqueness; source Theorems 2–3, 5–6)
Under (8.6) and Theorem 8.5, if $\mathcal{A}$ contains w_0 with $I[w_0] < \infty$, then I has a finite minimizer in $\mathcal{A}$. If $L = L(P, x)$ is independent of z and uniformly convex in P, this minimizer is unique. These statements apply to scalar and vector maps.

Expanded proof (the limit, and the midpoint deficit).

Choose $I[u_k] \downarrow \inf_{\mathcal{A}} I$, bounded above by $I[w_0] + 1$ and below by $-\beta|U|$. Coercivity and Poincaré give a weakly convergent subsequence, with limit in $\mathcal{A}$. Lower semicontinuity gives $I[u] \leq \inf_{\mathcal{A}} I$; the reverse inequality follows from feasibility. This proves existence. For two minimizers u, v, uniform convexity with constant $\theta > 0$ gives

$$I[(u + v)/2] + \frac{\theta}{8} \int_U |Du - Dv|^2 \leq \frac{1}{2}(I[u] + I[v]).$$

This follows by applying the supporting quadratic inequality at the midpoint and adding the two endpoint inequalities; the linear terms cancel. Since the midpoint is feasible, minimality forces $Du = Dv$. Poincaré for $u - v \in W_0^{1,q}$ then gives $u = v$. A merely convex dependence on P with arbitrary z dependence need not give uniqueness: the value variable may introduce several wells. □

The finite-competitor condition is stated explicitly here; otherwise a formal minimizer with energy $+\infty$ says nothing about the PDE.

为什么只对梯度要求凸

Rellich 给出 u_k 的强收敛，而 Du_k 通常只有弱收敛。支撑超平面把 Du_k 变成线性出现，正好能使用弱收敛；不能假定梯度有逐点收敛子列。

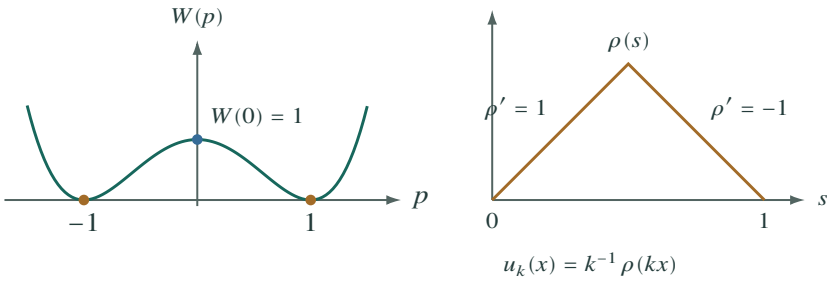


Diagram T8.2. Gradient oscillations test the need for convexity. The left graph is $W(p) = (p^2 - 1)^2$. A zero-trace triangular profile on $(0, 1)$ can have slopes $+1, -1$ while its amplitude tends to zero. Its energy is zero, whereas the limit profile has energy $W(0) = 1$. The right profile is one exact period, before multiplying its horizontal and vertical scales by $1/k$.

8.2.3 A Sobolev minimizer solves the Euler–Lagrange equation

Assume the additional growth bounds, with $q' = q/(q-1)$,

$$\begin{aligned} |L(P, z, x)| &\le C(1 + |P|^q + |z|^q), \\ |L_P| + |L_z| &\le C(1 + |P|^{q-1} + |z|^{q-1}). \end{aligned} \tag{8.7}$$

Theorem 8.7 (Weak Euler–Lagrange equation; source Theorems 4 and 7)
A minimizer $u \in \mathcal{A}$ satisfies

$$\int_U L_P(Du, u, x) : D\phi + L_z(Du, u, x) \cdot \phi = 0 \quad (\phi \in W_0^{1,q}(U; \mathbb{R}^m)).$$

Expanded proof (a difference quotient under an integrable bound).

For $|s| \le 1$, the fundamental theorem of calculus expresses $[L(Du + sD\phi, u + s\phi, x) - L(Du, u, x)]/s$ as an average of derivatives paired with $(D\phi, \phi)$. Its absolute value is bounded by

$$C(1 + |Du|^{q-1} + |u|^{q-1} + |D\phi|^{q-1} + |\phi|^{q-1})(|D\phi| + |\phi|),$$

which lies in L^1 by Hölder. Dominated convergence proves the first variation formula for every admissible ϕ. Both signs of s are allowed, so minimality makes the derivative zero. The coefficients $L_P(Du, u, x), L_z(Du, u, x)$ belong to $L^{q'}$, making this a continuous weak identity. Without the growth bound, differentiability of the energy on the whole Sobolev class has not been justified. □

8.2.4 Weak continuity of minors and polyconvexity

Lemma 8.8 (Weak determinant continuity; source §8.2.4)
If $q > n$ and $u_k \rightharpoonup u$ in $W^{1,q}(U;\mathbb{R}^n)$, then $\det Du_k \rightharpoonup \det Du$ in $L^{q/n}(U)$. More generally every r-minor converges weakly in $L^{q/r}$, $1 \le r \le n$.

Expanded proof (induction on minor order).

The gradients are bounded in L^q and Morrey plus compact embedding gives $u_k \to u$ uniformly after extraction. An r-minor is bounded in $L^{q/r}$ by the polynomial estimate $|M_r(Du_k)| \le C|Du_k|^r$. Its Laplace expansion along one fixed component can be written in divergence form, with that component undifferentiated and its $(r-1)$-cofactors inside the flux. The other derivative terms cancel by the same alternating-index argument as Piola. For a compactly supported smooth scalar test ψ, integration by parts therefore pairs $u_k^\alpha D\psi$, converging uniformly, with $(r-1)$-minors. By induction the latter converge weakly in $L^{q/(r-1)}$. Thus the r-minors converge in distributions; the base $r = 1$ is weak gradient convergence. Since $q/r > 1$, weak compactness and density of smooth tests identify every weak subsequential limit with $M_r(Du)$. This identifies the limit of the original sequence as well. The same divergence identity extends to Sobolev maps by local smoothing and Hölder, so it has not been used outside its legitimate class. □

The source uses the restricted polyconvex form

$$L(P,z,x) = F(P,\det P,z,x), \qquad (P,r) \mapsto F(P,r,z,x) \text{ convex.}$$

The standard broader definition permits all minors. We use the displayed restricted form, not an unmentioned change of definition in higher dimensions.

Theorem 8.9 (Polyconvex direct method; source Theorems 8–9)
If $q > n$, L is bounded below, and the smooth representation above holds, then I is weakly lower semicontinuous. If also (8.6) holds and a finite competitor exists, there is a minimizer.

Expanded proof (a supporting plane in an enlarged space).

Use Lemma 8.8 to get simultaneous weak convergence of Du_k in L^q and $\det Du_k$ in $L^{q/n}$, and uniform convergence of u_k along a subsequence. On a good set where $|u| + |Du|$ is bounded, apply the convex supporting inequality to F in the independent variables (P,r) at $(Du,\det Du)$. The

解方程与达到极小

一阶变分为零只给临界点。若 L 对 (P,z) 联合凸，则支撑不等式配合弱方程可推出全局极小；仅对 P 凸而对 z 非凸时，这个逆命题会失败。

非线性行列式的弱连续

一般弱收敛不能交换非线性映射。行列式的散度结构把一个因子变为强收敛的 u_k，剩余余子式逐阶归纳；$q > n$ 还保证 $L^{q/n}$ 反射。

two linear errors pair with the respective weakly convergent sequences; the coefficients converge uniformly and are bounded there. The argument of Theorem 8.5, including its truncated exhaustion of the domain, proves the inequality. Coercivity and weak closure finish the direct method. For example, $L(P) = |P|^4 + (\det P)^2$ for 2×2 matrices is of this form with $q = 4$. Polyconvexity does not mean convexity of the nonlinear map $P \mapsto (P, \det P)$. □

8.2.5 Local minimizers embedded in a field of extremals

Assume a smooth family $u(x, \lambda)$ solves the scalar Euler–Lagrange equation for each λ in an interval containing zero, with $u(x, 0) = u_0(x)$. Suppose L is convex in p. For a smooth $\vartheta : \overline{U} \to \mathbb{R}$ taking values in that interval, with $\vartheta = 0$ on ∂U, put $w(x) = u(x, \vartheta(x))$.

Theorem 8.10 (Field of extremals; source §8.2.5, Theorem 10)
For every such w, $I[w] \geq I[u_0]$. If u_λ is bounded away from zero near $\lambda = 0$, the inverse function theorem represents all sufficiently C^0-close smooth competitors with the same trace in this form.

Expanded proof (a calibration flux).

Here $D_x u$ denotes differentiation with λ fixed. The chain rule gives $Dw = D_x u + u_\lambda D\vartheta$ at $\lambda = \vartheta(x)$. Convexity bounds $L(Dw, w, x)$ below by $L(D_x u, w, x) + u_\lambda L_p(D_x u, w, x) \cdot D\vartheta$. Define the vector field

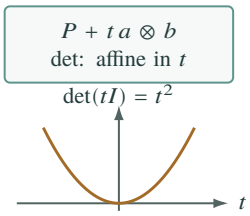
$$b_i(x) = \int_0^{\vartheta(x)} u_\lambda(x, s) L_{p_i}(D_x u(x, s), u(x, s), x)\, ds.$$

Differentiating the upper endpoint and using the Euler–Lagrange equation inside the integral gives

$$\begin{aligned} \operatorname{div} b &= u_\lambda L_p \cdot D\vartheta + \int_0^{\vartheta} (L_p \cdot D_x u_\lambda + L_z u_\lambda)\, ds \\ &= u_\lambda L_p \cdot D\vartheta + L(D_x u(x, \vartheta), w, x) - L(Du_0, u_0, x). \end{aligned}$$

Thus $I[w] \geq I[u_0] + \int_U \operatorname{div} b = I[u_0]$, since $b = 0$ on the boundary. The last assertion follows by a uniform local inverse in λ on the compact closed domain. Without that transversality, the theorem compares only competitors of the stated form. □

图 T8.M：秩一方向与全矩阵方向



对 2×2 矩阵，$t \mapsto \det(P + t\,a \otimes b)$ 是仿射函数，但 $t \mapsto \det(tI) = t^2$。秩一方向上的条件不等于整个矩阵空间中的凸性。

8.3 Regularity

8.3.1 Nonlinear difference quotients

Consider a scalar weak solution $u \in H^1(U)$ of

$$-\operatorname{div} DL(Du) = f, \qquad f \in L^2(U), \tag{8.8}$$

where $L \in C^2(\mathbb{R}^n)$ and

$$\theta|\xi|^2 \le D^2L(p)\xi\cdot\xi \le \Lambda|\xi|^2 \quad (p,\xi\in\mathbb{R}^n), \qquad 0<\theta\le\Lambda. \tag{8.9}$$

Subtracting the affine function $DL(0)\cdot p$ from L leaves the weak equation unchanged, because $\int_U DL(0)\cdot D\phi = 0$ for $\phi \in H^1_0$. We may therefore assume $DL(0)=0$, giving $|DL(p)| \le \Lambda|p|$.

Theorem 8.11 (Second derivatives; source §8.3.1, Theorem 1)
For $V \Subset U$, $u \in H^2(V)$ and

$$\|D^2u\|_{L^2(V)} \le C(\|f\|_{L^2(U)} + \|Du\|_{L^2(U)}).$$

If $u \in H^1_0(U)$ and ∂U is C^2, then $u \in H^2(U)$ and $\|u\|_{H^2(U)} \le C\|f\|_2$. Constants depend on θ, Λ, the domains and their indicated geometry.

Expanded proof (uniformly elliptic averaged Hessians).

Choose $V \Subset W \Subset U$ and a cutoff ζ supported in W, equal to one on V. For a small step in direction e_k let $\delta_h v = (v(x+he_k)-v(x))/h$ and use the legal test $\phi = -\delta_{-h}(\zeta^2\delta_h u)$. It belongs to $H^1_0(U)$ because its support stays away from the boundary. Discrete integration by parts gives

$$\int_U \delta_h DL(Du)\cdot D(\zeta^2\delta_h u) = -\int_U f\,\delta_{-h}(\zeta^2\delta_h u).$$

The exact nonlinear linearization is

$$\delta_h DL(Du) = A_h\delta_h Du, \quad A_h = \int_0^1 D^2L(Du + s(Du(x+he_k)-Du))\,ds.$$

Hence $\theta I \le A_h \le \Lambda I$ independently of h. Expanding the cutoff derivative, the principal integral bounds $\theta\|\zeta\delta_h Du\|_2^2$. Its cross term costs $\varepsilon\|\zeta\delta_h Du\|_2^2 + C_\varepsilon\|\delta_h u\|_2^2$. The forcing is controlled without differentiating f:

$$\begin{aligned}\left|\int f\,\delta_{-h}(\zeta^2\delta_h u)\right| &\le C\|f\|_2\|D(\zeta^2\delta_h u)\|_2\\ &\le \varepsilon\|\zeta\delta_h Du\|_2^2 + C_\varepsilon(\|f\|_2^2 + \|Du\|_2^2).\end{aligned}$$

Use $\|\delta_h u\|_2 \le \|Du\|_2$ on a slightly larger set and absorb with $2\varepsilon<\theta$. The difference-quotient criterion from §5.8 then gives the interior estimate. □

Expanded proof (boundary and normal estimates).

This expands the boundary argument left to the reader in the source. In a boundary chart write $x = \Phi(y) = (y', y_n + \gamma(y'))$, $v = u \circ \Phi$. Its Jacobian is one and $v = 0$ on $y_n = 0$. The transformed flux is

$$a(y,p) = D\Phi^{-1}\, DL(D\Phi^{-T}p), \qquad -\partial_{y_i} a_i(y, Dv) = \tilde{f}.$$

Bounded $D\Phi, D\Phi^{-1}$ preserve uniform monotonicity in p; bounded $D^2\Phi$ gives $|D_y a(y,p)| \le C|p|$, since $DL(0) = 0$. Tangential difference quotients now work on the half-chart. Their tests have zero trace because v and its tangential translates have zero trace. Besides the averaged Hessian term, differencing a produces a coefficient error bounded by $C|Dv|$, which is absorbed as above. Thus all $v_{i\alpha}$, $\alpha < n$, are in L^2, with bounds by $\|\tilde{f}\|_2 + \|Dv\|_2$.

To recover v_{nn} without assuming it exists, let $w = a_n(y, Dv)$. The known tangential derivatives and the flux chain rule give $\partial_\alpha a_i(y, Dv) \in L^2$. The equation gives $\partial_n w = -\tilde{f} - \sum_{\alpha<n} \partial_\alpha a_\alpha \in L^2$. For fixed (y, p'), $p_n \mapsto a_n(y, p', p_n)$ is strictly increasing with derivative at least $c > 0$ and at most C; it maps $\mathbb{R}$ onto $\mathbb{R}$. Its inverse $p_n = \Psi(y, p', w)$ satisfies

$$|\Psi_w| + |\Psi_{p'}| \le C, \qquad |\Psi_y| \le C(|p'| + |\Psi|).$$

Apply normal difference quotients to $v_n = \Psi(y, D'v, w)$. The L^2 bounds for the normal derivatives of $D'v$ and w give an L^2 bound for the normal difference quotients of v_n. The criterion proves $v_{nn} \in L^2$; this step avoids a circular classical expansion of an equation before H^2 has been established. Cutoffs, a finite chart cover and the interior estimate yield $\|u\|_{H^2} \le C(\|f\|_2 + \|u\|_{H^1})$. Finally testing the original equation by u, monotonicity and Poincaré give

$$\theta\|Du\|_2^2 \le \int_U DL(Du)\cdot Du = (f,u) \le C\|f\|_2\|Du\|_2.$$

This removes the lower-order norm and proves the global estimate. □

平均 Hessian 保留椭圆性

A_h 是沿两点梯度连线的 Hessian 平均；一致上下界在平均后不变。不能只在某个未经控制的中间点套用局部凸性。

8.3.2 Why further smoothness needs a new theorem

For $f = 0$, interior H^2 and the Lipschitz chain rule imply that $v_k = u_{x_k}$ weakly solves

$$-\partial_i(A^{ij}(x)\partial_j v_k) = 0, \qquad A^{ij}(x) = L_{p_i p_j}(Du(x)).$$

These coefficients are only bounded measurable at this stage. The smooth coefficient estimates of Chapter 6 cannot be applied to them. The scalar De Giorgi–Nash–Moser theorem, used here as an external regularity result, states that H^1_{loc} solutions of a uniformly elliptic divergence equation with bounded measurable coefficients are locally Hölder continuous. It gives $Du \in C^{0,\alpha}_{\mathrm{loc}}$. For smooth L, A is then Hölder. The external Schauder estimates for linear divergence equations improve v_k to $C^{1,\alpha}_{\mathrm{loc}}$; iteration improves coefficient and solution regularity together, yielding $u \in C^\infty(U)$. This is a conditional bootstrap using those two named theorems, not a proof of either deep estimate. For systems the scalar Hölder theorem is unavailable in general; smoothness cannot be asserted by the same argument.

法向导数的非循环恢复

先证明切向二阶导数，再由方程得到法向通量的弱导数，最后利用 $p_n \mapsto a_n$ 的一致单调可逆性恢复 u_{nn}。不能在已知 $u \in H^1$ 时直接展开二阶链式法则。

8.4 Constraints

8.4.1 Integral constraints and a scalar multiplier

On connected U, minimize $I[u] = \frac{1}{2}\int_U |Du|^2$ over

$$\mathcal{A} = \{u \in H^1_0(U) : J[u] = 0\}, \qquad J[u] = \int_U G(u).$$

Assume $g = G'$ satisfies $|g(z)| \le C(1 + |z|)$. Then $|G(z)| \le C(1 + z^2)$, so the constraint is well defined.

> **Theorem 8.12** (Constrained existence; source §8.4.1, Theorem 1)
> *If $\mathcal{A} \neq \emptyset$, it contains an energy minimizer.*

Expanded proof (compactness preserves the nonlinear integral).

An energy-bounded minimizing sequence is bounded in H^1_0. After extraction $u_k \rightharpoonup u$ in H^1_0 and $u_k \to u$ in L^2. The mean value theorem and Cauchy–Schwarz give

$$|J[u_k] - J[u]| \le C\|u_k - u\|_2(|U|^{1/2} + \|u_k\|_2 + \|u\|_2) \longrightarrow 0.$$

Thus u is feasible, and weak lower semicontinuity of the Dirichlet norm makes it minimizing. Every member of this particular class has finite energy. □

线性化不意味着光滑系数

$A(x) = D^2L(Du(x))$ 的正则性由未知梯度决定。先得到梯度的 Hölder 连续性，才能启动 Schauder 提升；多次形式求导不能绕过这一步。

Theorem 8.13 (Lagrange multiplier; source Theorem 2)
For a minimizer u there is $\lambda \in \mathbb{R}$ such that

$$\int_U Du \cdot D\phi = \lambda \int_U g(u)\phi \quad (\phi \in H_0^1(U)). \tag{8.10}$$

Expanded proof (an admissible curve and the degenerate case).

First suppose $g(u) \not\equiv 0$. By density choose $\psi \in H_0^1$ with $b = \int_U g(u)\psi \neq 0$. For any $\phi \in H_0^1$, set $j(s,t) = J[u + s\phi + t\psi]$. Its derivatives are continuous by the linear growth bound on g and dominated convergence on this finite parameter plane. Since $j_t(0,0) = b$, the ordinary implicit function theorem supplies $t = \eta(s)$ with $j(s,\eta(s)) = 0$, $\eta(0) = 0$, and $\eta'(0) = -\int g(u)\phi/b$. Minimality on this curve gives $\int Du \cdot D\phi + \eta'(0) \int Du \cdot D\psi = 0$. Thus (8.10) holds with $\lambda = (\int Du \cdot D\psi)/b$, independent of ϕ.

If $g(u) = 0$ a.e., the chain rule with truncation gives $G(u) \in W^{1,1}$ and $D(G(u)) = g(u)Du = 0$. The product is integrable by Cauchy–Schwarz. Connectedness makes $G(u)$ constant, and $J[u] = 0$ makes it zero. To identify its boundary value, approximate u in H_0^1 by compactly supported smooth functions: the linear growth of g implies convergence of their compositions in $W^{1,1}$, so $G(u) - G(0)$ has zero trace. Hence $G(0) = 0$ and zero is feasible. Minimality gives $I[u] = 0$, whence $u = 0$. Both sides of (8.10) vanish for every λ. This final conclusion is special to this energy and constraint, not a universal multiplier theorem without a constraint qualification. □

退化约束不能直接除法

只有 $J'[u] \neq 0$ 时才能用隐函数定理并除以 $\int g(u)\psi$。本例退化时另证 $u = 0$；一般约束问题不保证退化点存在普通 Lagrange 乘子。

8.4.2 The obstacle problem and a variational inequality

Let $f \in L^2(U)$, let h be a smooth obstacle, and assume $K = \{w \in H_0^1(U) : w \geq h \text{ a.e.}\} \neq \emptyset$. Minimize $I[w] = \frac{1}{2}\|Dw\|_2^2 - (f, w)$.

Theorem 8.14 (Obstacle minimizer; source §8.4.2, Theorems 3–4)
There is a unique minimizer $u \in K$, characterized equivalently by

$$\int_U Du \cdot D(w - u) \geq \int_U f(w - u) \qquad (w \in K). \tag{8.11}$$

Expanded proof (one-sided variations and the converse).

Young and Poincaré imply $I[w] \geq \frac{1}{4}\|Dw\|_2^2 - C\|f\|_2^2$. Weak limits of minimizing sequences stay in K: strong L^2 convergence has an a.e. convergent subsequence preserving $w \geq h$. The direct method gives existence. The midpoint identity gives strict convexity on H_0^1, hence uniqueness. For $w \in K$, the segment $u + s(w - u) \in K$ for $0 \leq s \leq 1$. Its

right energy derivative at zero is nonnegative, giving (8.11). Conversely, expand the quadratic energy exactly:

$$I[w] - I[u] = \int_U Du \cdot D(w-u) - f(w-u) + \tfrac{1}{2}\|D(w-u)\|_2^2 \geq 0.$$

This proves equivalence, including sufficiency of the inequality. □

For every nonnegative $\phi \in C_c^\infty(U)$, $u + \phi \in K$ gives $\mu = -\Delta u - f \geq 0$ in distributions. If u is continuous, then $O = \{u > h\}$ is open; on a compact subset of O both signs of a small variation are allowed, so $-\Delta u = f$ there. If also $u \in H^2_{\mathrm{loc}}$, then μ is a function and

$$u - h \geq 0, \quad \mu \geq 0, \quad (u-h)\mu = 0 \quad \text{a.e.}$$

Interior continuity and $C^{1,1}_{\mathrm{loc}}$ regularity for smooth obstacle and forcing require external obstacle regularity theory; they are not proved by the direct method above. No global $W^{2,\infty}$ estimate is claimed without specifying its boundary compatibility hypotheses. The free boundary is $\partial\{u > h\} \cap U$.

接触集与反力

约束只允许向上变化，故反力 $\mu = -\Delta u - f$ 非负。在非接触区可双向变化，反力为零。可行集、反力和互补条件要使用同一符号约定。

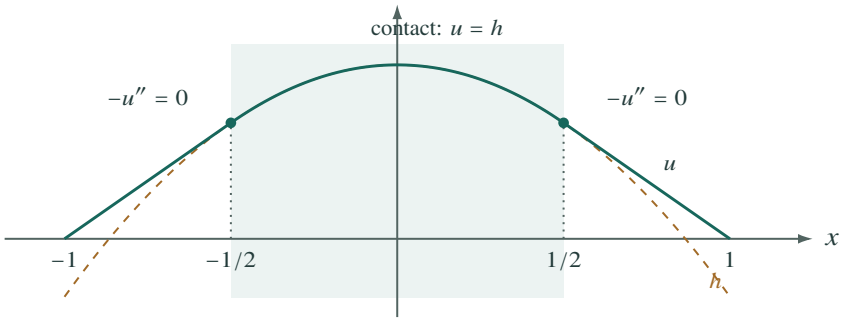


Diagram T8.3. An exact obstacle solution with two free-boundary points. On $(-1,1)$ take $f = 0$ and $h(x) = 1 - 4x^2/3$. The minimizer is $u = h$ on $|x| \leq 1/2$, and $u = 4(1-|x|)/3$ outside. Values and first derivatives match at $\pm 1/2$, so $-u'' = 8/3$ on the contact interval and zero elsewhere, with no point masses.

8.4.3 Pointwise constraints: harmonic maps

Consider $\mathcal{A} = \{u \in H^1(U;\mathbb{R}^m) : u - g \in H^1_0,\ |u| = 1 \text{ a.e.}\}$, and $I[u] = \frac{1}{2}\int_U |Du|^2$. A minimizing sequence is bounded in H^1 ($|u| = 1$ fixes its L^2 norm). A strong L^2 subsequence preserves the sphere constraint a.e.; the trace condition is weakly closed, so a minimizer exists if the class is nonempty.

Theorem 8.15 (Harmonic map equation; source §8.4.3, Theorem 5)
A minimizer satisfies

$$\int_U Du : D\phi = \int_U |Du|^2 u \cdot \phi \quad (\phi \in H_0^1(U;\mathbb{R}^m) \cap L^\infty(U;\mathbb{R}^m)). \tag{8.12}$$

Expanded proof (normalization generates tangent variations).

For bounded ϕ, $u_s = (u + s\phi)/|u + s\phi|$ is well defined for $|s|\|\phi\|_\infty < 1/2$. Sobolev composition and traces show $u_s \in \mathcal{A}$. Its derivative at zero in H^1 is $\dot{u}_0 = \phi - (u \cdot \phi)u$. To justify this derivative, derivatives of the normalization map are uniformly bounded on $|u + s\phi| \geq 1/2$; its spatial derivative and parameter derivative are dominated in L^2 by $C(|D\phi| + |Du|\,\|\phi\|_\infty)$. Thus differentiating the energy is legitimate. Since $u \cdot \partial_i u = 0$ a.e.,

$$Du : D((u \cdot \phi)u) = |Du|^2 (u \cdot \phi).$$

The vanishing derivative is precisely (8.12). The multiplier is the function $|Du|^2 \in L^1$, explaining why bounded variations appear in the weak formulation. □

8.4.4 Divergence constraints and pressure

For $f \in L^2(U;\mathbb{R}^n)$ minimize $I[v] = \frac{1}{2}\int |Dv|^2 - \int f \cdot v$ on the closed linear space $V_\sigma = \{v \in H_0^1(U;\mathbb{R}^n) : \operatorname{div} v = 0\}$. Coercivity and strict convexity give a unique u, with $\ell(v) = \int Du : Dv - f \cdot v = 0$ for $v \in V_\sigma$. To recover pressure we use the following external divergence theorem: on a bounded connected smooth domain there is a bounded right inverse

$$\mathcal{B} : L_0^2(U) \to H_0^1(U;\mathbb{R}^n), \quad \operatorname{div} \mathcal{B}q = q, \quad \|\mathcal{B}q\|_{H^1} \leq C\|q\|_2.$$

Here L_0^2 denotes zero mean. This is the divergence-solving estimate also invoked without proof in the source, often realized by a Bogovskii operator. Its construction is not proved here.

约束曲率进入方程

球面约束的切向投影是 $\phi - (u \cdot \phi)u$。对 $|u|^2 = 1$ 求导给 $u \cdot u_i = 0$；再次求导产生 $|Du|^2 u$，这里乘子是函数而非一个常数。

Theorem 8.16 (Pressure multiplier; source §8.4.4, Theorem 6)
There is a unique mean-zero $p \in L^2(U)$ satisfying

$$\int_U Du : Dv = \int_U f \cdot v + p \operatorname{div} v \quad (v \in H_0^1(U; \mathbb{R}^n)).$$

Thus $-\Delta u + Dp = f$, $\operatorname{div} u = 0$ in distributions. The global conclusion uses the external right inverse just stated; the source proves a local pressure statement.

Expanded proof (factoring a functional through divergence).

The functional $q \mapsto \ell(\mathcal{B}q)$ is bounded on L_0^2. Riesz representation gives mean-zero p with $\ell(\mathcal{B}q) = \int pq$. For any $v \in H_0^1$, $q = \operatorname{div} v$ has zero mean and $v - \mathcal{B}q \in V_\sigma$. Hence $\ell(v) = \ell(\mathcal{B}q) = \int p \operatorname{div} v$. Surjectivity of divergence gives uniqueness in L_0^2 and $\|p\|_2 \le C(\|Du\|_2 + \|f\|_2)$. Without mean normalization constants are undetermined, since $\int \operatorname{div} v = 0$. This argument needs connectedness and the right inverse, not a claim that every curl-free field is globally a gradient. □

8.4.5 Nonlinear incompressibility

In three-dimensional elasticity, u is the deformation map (the new position), not the displacement $u - x$. Local incompressibility is $\det Du = 1$. For an actual displacement $v = u - x$ it is $\det(I + Dv) = 1$, whose first-order linearization is $\operatorname{div} v = 0$.

Theorem 8.17 (Determinant constraint; source Theorem 7)
For $q > 3$, let $L(P, x)$ be smooth, convex in P, and satisfy (8.6). If

$$\mathcal{A} = \{u \in g + W_0^{1,q}(U; \mathbb{R}^3) : \det Du = 1 \text{ a.e.}\}$$

has a finite-energy member, then it has an energy minimizer.

Expanded proof (weak closure of the determinant condition).

Coercivity bounds a minimizing sequence in $W^{1,q}$. Its weak limit keeps the trace, and Lemma 8.8 gives $\det Du_k \rightharpoonup \det Du$ in $L^{q/3}$. Since every determinant equals one, the limit does too. Convex lower semicontinuity completes existence. The feasible set is not convex, so convexity of L alone supplies no uniqueness assertion here. The determinant condition is local and does not by itself prove global injectivity or prevent self-interpenetration. □

压力只确定到常数

测试速度具有零边界，故散度积分为零；压力加常数不改变弱式。右逆估计将消失在无散度空间上的泛函表示为 $p \operatorname{div} v$，并给出压力范数控制。

8.5 Critical Points

8.5.1 Palais–Smale compactness and a deformation

Let H be a real Hilbert space and $I \in C^1(H;\mathbb{R})$. Write $\nabla I(u) \in H$ for the Riesz representative of $DI(u)$; the source writes this representative as $I'[u]$. Assume ∇I is Lipschitz on bounded subsets, the source's class $\mathscr{C}$. A critical point has $\nabla I(u) = 0$. The Palais–Smale condition says that every sequence with bounded $I(u_k)$ and $\nabla I(u_k) \to 0$ has a strongly convergent subsequence in H. Boundedness of the sequence alone would not suffice in infinite dimension.

Theorem 8.18 (Deformation theorem; source §8.5.1, Theorem 1)
Suppose $I \in \mathscr{C}$ satisfies Palais–Smale and has no critical point at level c. For sufficiently small $\varepsilon > 0$ there are $0 < \delta < \varepsilon$ and a continuous flow $\eta : [0,1] \times H \to H$ with $\eta(0,u) = u$, fixing points outside $\{|I - c| < \varepsilon\}$, not increasing I, and satisfying

$$\eta(1, \{I \le c + \delta\}) \subset \{I \le c - \delta\}.$$

Expanded proof (a bounded descending vector field).

There exist $\varepsilon_0 > 0, 0 < \sigma \le 1$ such that $\|\nabla I(u)\| \ge \sigma$ whenever $|I(u) - c| \le \varepsilon_0$. Otherwise choose $|I(u_k) - c| \le 1/k$ and $\|\nabla I(u_k)\| \le 1/k$; Palais–Smale and continuity give a critical point at c, a contradiction. Choose $0 < \varepsilon \le \varepsilon_0$, a Lipschitz scalar cutoff χ equal to one on $[c - \varepsilon/2, c + \varepsilon/2]$ and zero outside $(c - \varepsilon, c + \varepsilon)$, and define

$$V(u) = -\chi(I(u)) \frac{\nabla I(u)}{\max\{1, \|\nabla I(u)\|\}}.$$

It is Lipschitz on bounded sets and has norm at most one. The Banach-space ODE $\dot\eta = V(\eta)$ has unique global solutions: Picard iteration gives local existence, and bounded speed makes trajectories Cauchy at a finite endpoint, allowing continuation. Continuous dependence gives the required continuous flow, and $V = 0$ outside the energy band fixes those points. The Hilbert chain rule gives

$$\frac{d}{dt} I(\eta) = -\chi(I(\eta)) \frac{\|\nabla I(\eta)\|^2}{\max\{1, \|\nabla I(\eta)\|\}} \le 0.$$

Choose $0 < \delta < \varepsilon/2$ with $2\delta < \sigma^2$. An orbit starting at energy at most $c + \delta$ is already below $c - \delta$, exits the narrow band downward, or stays in it for time one. In the last case $\chi = 1$ and the derivative is at most $-\sigma^2$, so it too ends below $c - \delta$. This proves every asserted property. □

8.5.2 The mountain pass level

Theorem 8.19 (Mountain pass; source Theorem 2)
Let $I \in \mathscr{C}$ satisfy Palais–Smale, $I(0) = 0$, and suppose $I(u) \geq a > 0$ when $\|u\| = r$, while some e satisfies $\|e\| > r$, $I(e) \leq 0$. For

$$\Gamma = \{\gamma \in C([0,1]; H) : \gamma(0) = 0,\ \gamma(1) = e\},$$
$$c = \inf_{\gamma \in \Gamma} \max_{s \in [0,1]} I(\gamma(s)),$$

there is $u \in H$ with $I(u) = c \geq a$ and $\nabla I(u) = 0$.

Expanded proof (deforming a nearly optimal path).

Every path crosses the radius-r sphere by continuity of its norm, so $c \geq a$. The straight path shows c is finite. If c were not critical, use Theorem 8.18 with $\varepsilon < a/2$. Choose a path with maximum at most $c + \delta$. The deformed path has maximum at most $c - \delta$; its endpoints remain $0, e$, because their energies lie below $c - \varepsilon$. It still belongs to Γ, contradicting the definition of c. The conclusion is existence at a minimax level; it does not assert nondegeneracy, a particular Morse index, or uniqueness of the critical point. □

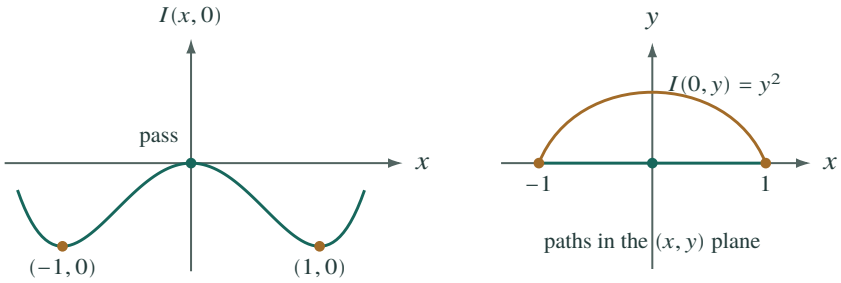


Diagram T8.4. A minimax path must cross an energy barrier. This is an exact two-dimensional model, $I(x,y) = (x^2 - 1)^2 + y^2 - 1$, connecting $(1,0)$ to $(-1,0)$. Both endpoints have energy -1 and every path crosses $x = 0$, where $I \geq 0$. The horizontal path attains maximum zero at the saddle $(0,0)$. Translation and addition of a constant put the first endpoint at the theorem's 0 with energy zero.

Palais–Smale 的用途

若临界层附近不存在梯度下界，就能构造能量趋于 c、梯度趋零的序列。PS 将它变为强收敛子列，从而得到被假设排除的临界点。

8.5.3 A nonzero solution of a semilinear Dirichlet equation

Let $n \geq 3$, $1 < p < (n+2)/(n-2)$, $f \in C^1(\mathbb{R})$, and $F(s) = \int_0^s f(\tau)\,d\tau$. Assume for some $a, A > 0$ and $\mu > 2$,

$$\begin{aligned} |f(s)| &\leq C(1+|s|^p), \quad |f'(s)| \leq C(1+|s|^{p-1}), \\ a|s|^{p+1} &\leq F(s) \leq A|s|^{p+1}, \qquad \mu F(s) \leq s f(s). \end{aligned} \tag{8.13}$$

These are the source assumptions with $\mu = 1/\gamma > 2$.

Theorem 8.20 (Semilinear application; source §8.5.2, Theorem 3)
The problem $-\Delta u = f(u)$, $u \in H_0^1(U)$, has a nonzero weak solution.

Expanded proof (differentiability and compactness).

Use $H = H_0^1(U)$ with $\|u\|_H = \|Du\|_2$ and define $I(u) = \frac{1}{2}\|u\|_H^2 - \int F(u)$. Sobolev embedding into L^{p+1} makes I finite. The mean value bound on f' gives

$$\|f(u) - f(v)\|_{L^{(p+1)/p}} \leq C(1 + \|u\|_{p+1}^{p-1} + \|v\|_{p+1}^{p-1})\|u - v\|_{p+1}.$$

Together with the quadratic Taylor remainder for F, this proves Fréchet differentiability and a gradient Lipschitz on bounded H sets. If $K : H^{-1} \to H_0^1$ is the inverse Dirichlet Laplacian, Riesz representation gives $\nabla I(u) = u - Kf(u)$. The nonlinear functional $v \mapsto \int f(u)v$ is in H^{-1} by Hölder and the L^{p+1} embedding; no pointwise identification of the Hilbert gradient with $-\Delta u - f(u)$ is made.

For a Palais–Smale sequence, write $\epsilon_k = \|\nabla I(u_k)\| \to 0$. The superquadratic inequality yields

$$\left(\tfrac{1}{2} - \tfrac{1}{\mu}\right)\|u_k\|_H^2 \leq I(u_k) - \tfrac{1}{\mu} DI(u_k)[u_k] \leq C + \tfrac{1}{\mu}\epsilon_k \|u_k\|_H.$$

Thus u_k is bounded in H. Since $p+1 < 2^* = 2n/(n-2)$, a subsequence converges strongly in L^{p+1}; the preceding estimate gives $f(u_k) \to f(u)$ in H^{-1}. Then $u_k = Kf(u_k) + o_H(1)$ converges strongly in H. This verifies Palais–Smale, including its strong compactness conclusion.

Finally $I(u) \geq \frac{1}{2}\|u\|_H^2 - C\|u\|_H^{p+1}$ is positive on a small sphere, since $p + 1 > 2$. For any nonzero $\phi \in C_c^\infty(U)$,

$$I(t\phi) \leq \tfrac{1}{2}t^2\|\phi\|_H^2 - at^{p+1}\|\phi\|_{p+1}^{p+1} \longrightarrow -\infty.$$

The mountain pass theorem supplies a positive-energy critical point, which is nonzero because $I(0) = 0$. Its weak equation follows from $DI(u)[v] = 0$. In dimensions one and two the same proof works for finite $p > 1$ using the corresponding finite-exponent embeddings. □

8.6 Invariance, Noether's Theorem

8.6.1 Domain variations and a sign convention for the current

Let Φ_s be smooth local diffeomorphisms near $\overline{U}$, with $\Phi_0(x) = x$, and w_s smooth functions with $w_0 = u$. Write $X = \partial_s \Phi_s|_0$ and $m = \partial_s w_s|_0$. The source's invariance convention is

$$\int_V L(Dw_s, w_s, x)\,dx = \int_{\Phi_s(V)} L(Du, u, x)\,dx \tag{8.14}$$

for every relatively compact subdomain V, not just the whole U. The variations need not preserve a particular prescribed boundary value; Noether's conclusion is a local identity for smooth solutions.

Theorem 8.21 (Noether identity; source §8.6.2, Theorem 1)
Under (8.14)*,*

$$\operatorname{div}(mL_p - LX) = m(\operatorname{div} L_p - L_z). \tag{8.15}$$

Thus $J = mL_p - LX$ is divergence free on Euler–Lagrange solutions.

Expanded proof (local invariance and the current).

The derivative of the left side is $\int_V L_p \cdot Dm + L_z m$. For the right side, changing variables $y = \Phi_s(x)$ and using $\frac{d}{ds} \det D\Phi_s|_0 = \operatorname{div} X$ gives $\int_V \operatorname{div}(LX)$; here differentiation of the composed $L(Du, u, x)$ is a total spatial derivative. Since this holds for every V, the smooth integrands coincide pointwise: $L_p \cdot Dm + L_z m = \operatorname{div}(LX)$. Expanding $\operatorname{div}(mL_p)$ proves (8.15). Invariance on only one fixed domain would yield only an integral identity, not this pointwise conservation law. □

8.6.2 Translations, scaling, and monotonicity

If $L = L(p, z)$ is independent of x, translations give $X = e_k$, $m = u_k$. Therefore the stress tensor

$$T_{ik} = L_{p_i} u_k - L\delta_{ik} \quad \text{satisfies} \quad \partial_i T_{ik} = 0.$$

For L with explicit x dependence, direct differentiation instead gives $\partial_i T_{ik} = -L_{x_k}$ on a solution. Explicit inhomogeneity is a source term, not a broken algebraic identity. For $I[u] = \int |Du|^p$, the scaling

$$\Phi_s(x) = e^s x, \quad w_s(x) = e^{(n-p)s/p} u(e^s x)$$

obeys (8.14): the gradient contributes e^{ns}, which cancels the change-of-variables Jacobian. Thus $X = x$ and $m = x \cdot Du + (n-p)u/p$.

严格次临界保证紧性

$p + 1 < 2^*$ 不仅保证能量可积，更让 u_k 在 L^{p+1} 强收敛。临界指数处连续嵌入仍在，但紧性丢失，不能照搬这个 PS 证明。

守恒流的符号约定

本书用主动变化 $w_s(x) = u(\Phi_s(x))$，对应 $J = mL_p - LX$。若换成逆映射的被动约定，公式会换号；不能混用两种定义。

Theorem 8.22 (Scaled energy monotonicity; source §8.6.2, Example 2)
For a smooth p-harmonic critical point for which the preceding calculation is classical, or a weak local minimizer of $\int |Du|^p$, $p > 1$, *set* $E(r) = \int_{B_r} |Du|^p$ *with* $B_R \Subset U$. *For a.e.* $0 < r < R$,

$$\frac{d}{dr}\left(r^{p-n}E(r)\right) = pr^{p-n}\int_{\partial B_r} |Du|^{p-2}u_r^2\, dS \geq 0. \tag{8.16}$$

The integrand is defined as zero where $Du = 0$.

Expanded proof (from an inner variation to a radial flux).

For smooth solutions the Noether current above has zero boundary flux on B_r. The term containing u integrates to zero because $\int_{\partial B_r} u|Du|^{p-2}u_r = \int_{B_r} |Du|^p$, by the p-Laplace equation. Rearrangement gives

$$(n-p)E(r) = r\int_{\partial B_r} (|Du|^p - p|Du|^{p-2}u_r^2).$$

Since $E'(r) = \int_{\partial B_r} |Du|^p$ a.e., differentiation gives (8.16) with the displayed power of r. For a weak local minimizer avoid differentiating Du: take compactly supported inner variations $u \circ (\mathrm{id} + sY)$. After changing variables, only the matrix and Jacobian are differentiated; the derivative is dominated by $C|Du|^p$ and yields

$$\int_U |Du|^p \operatorname{div} Y - p|Du|^{p-2}Du \cdot (DY)Du = 0.$$

Use radial cutoffs $Y(x) = \chi(|x|)x$ approaching the indicator of B_r. Coarea and Lebesgue differentiation give the same radial identity for a.e. r. This extends the proof without assuming L is C^2 at zero gradient when $1 < p < 2$. The scaled energy has its natural nondecreasing representative. □

8.6.3 Time translation and wave energy

For the action with kinetic minus potential density

$$\mathscr{L}(D_x u, u_t, u) = \tfrac{1}{2}u_t^2 - \tfrac{1}{2}|D_x u|^2 - F(u),$$

the Euler–Lagrange equation is $u_{tt} - \Delta u + f(u) = 0$, $F' = f$. Time translation gives $m = u_t$ and the space–time current $(-u_t D_x u, e)$, where

$$e = \tfrac{1}{2}(u_t^2 + |D_x u|^2) + F(u).$$

Hence $e_t - \operatorname{div}_x(u_t D_x u) = 0$. For compact spatial support, or zero boundary flux, $\int e$ is constant. A signed F still gives conservation; nonnegativity is an additional requirement if this energy is to control norms.

单调量含有尺度因子

单调的是 $r^{p-n}\int_{B_r}|Du|^p$。右侧仅含径向导数的非负平方；这个恒等式比“积分区域增大所以能量增大”包含更多信息。

For the linear wave action, simultaneous scaling of (x,t) has multiplier $m = tu_t + x\cdot Du + (n-1)u/2$ and gives $P_t - \operatorname{div} Q = 0$, with

$$P = te_0 + (x\cdot Du)u_t + \tfrac{n-1}{2}uu_t,$$
$$Q = mDu + \tfrac{1}{2}(u_t^2 - |Du|^2)x, \qquad e_0 = \tfrac{1}{2}(u_t^2 + |Du|^2).$$

The exponent $(n-1)/2$ comes from the space–time dimension $n+1$.

8.6.4 Conformal multiplier and local energy decay

The source motivates a further multiplier by hyperbolic inversion $(x,t) \mapsto (x,t)/(|x|^2 - t^2)$ and the wave Kelvin transform

$$u \mapsto |\,|x|^2 - t^2\,|^{-(n-1)/2} u\left(\frac{x}{|x|^2-t^2}, \frac{t}{|x|^2-t^2}\right),$$

away from its singular cone. Conjugating a time translation suggests

$$M = (r^2+t^2)u_t + 2t\,x\cdot Du + (n-1)tu, \qquad r = |x|.$$

We do not assume the action is invariant under this transformation. Instead the useful conclusion follows from the direct multiplier identity

$$C_t - \operatorname{div} R = M(u_{tt} - \Delta u), \tag{8.17}$$

where

$$C = (r^2+t^2)e_0 + 2t(x\cdot Du)u_t + (n-1)tuu_t - \tfrac{n-1}{2}u^2,$$
$$R = MDu + t(u_t^2 - |Du|^2)x.$$

Expanded proof (conformal current and its sign).

Expand $M(u_{tt} - \Delta u)$, integrating each second derivative once by the product rule. The second-derivative terms give $C_t - \operatorname{div} R$. The remaining coefficients of u_t^2, $|Du|^2$ and uu_t cancel using $\partial_t(r^2+t^2) = 2t$, $D(r^2+t^2) = 2x$ and $\operatorname{div} x = n$, proving (8.17) off shell. For a more useful sign, put $\alpha = (n-1)/2$ and $u_r = Du\cdot x/r$. Direct expansion gives $C = S - \frac{\alpha}{2}\operatorname{div} Z$, with

$$\begin{aligned} S = {} & \frac{(t+r)^2}{4}(u_t + u_r + \alpha u/r)^2 \\ & + \frac{(t-r)^2}{4}(u_t - u_r - \alpha u/r)^2 \\ & + \frac{r^2+t^2}{2}\left(|Du|^2 - u_r^2 + \frac{(n-3)(n-1)}{4r^2}u^2\right), \\ Z = {} & \frac{r^2+t^2}{r^2}u^2 x. \end{aligned}$$

Indeed expanding the two squares produces the cross term $2\alpha tuu_t$ and an extra $\alpha(r^2+t^2)uu_r/r$. The latter and the remaining u^2 terms are exactly $\frac{\alpha}{2}\operatorname{div} Z$, with $\operatorname{div} Z = 2(r^2+t^2)uu_r/r + [n + (n-2)t^2/r^2]u^2$. For $n \geq 3$, every term of S is nonnegative. The pointwise completion is used away from $r = 0$; the exterior application below has the origin inside the excluded obstacle. □

Theorem 8.23 (Exterior local energy decay; source application, pp.487–489)
Let $n \geq 3$ and let a smooth bounded obstacle O contain a neighborhood of the origin and be star-shaped about it. A smooth solution of the wave equation on $\Omega = \mathbb{R}^n \setminus \overline{O}$, with zero Dirichlet boundary values and compatible compactly supported initial data, satisfies

$$\int_{B_R \setminus \overline{O}} (u_t^2 + |Du|^2)\, dx \leq C_R t^{-2} \quad (t \geq 2R), \qquad \overline{O} \subset B_R.$$

Expanded proof (the obstacle flux and the radial correction).

Finite propagation keeps all fields compactly supported at each finite time. The outer normal ν to Ω points into O. On ∂O, $u = u_t = 0$ and $Du = (\partial_\nu u)\nu$, so $R \cdot \nu = t|Du|^2 x \cdot \nu \leq 0$ by star-shapedness. Integrating (8.17) gives $\int_\Omega C(t) \leq \int_\Omega C(0)$. The divergence correction has zero total flux, since $u = 0$ on the obstacle and is compactly supported. Thus $\int_\Omega S(t) \leq C$. For $r \leq R$, $t \geq 2R$, both $(t \pm r)^2 \geq t^2/4$; hence

$$\int_{B_R \setminus \overline{O}} u_t^2 + (u_r + \alpha u/r)^2 + (|Du|^2 - u_r^2) + (n-3)u^2/r^2 \leq C_R t^{-2}.$$

To extract u_r^2, integrate the radial cross term on this bounded region:

$$\int (u_r + \alpha u/r)^2 = \int u_r^2 + \alpha \int_{\partial B_R} u^2/R + \frac{(n-1)(3-n)}{4} \int u^2/r^2.$$

There is no inner boundary contribution because $u = 0$ there. For $n = 3$ the last coefficient vanishes; for $n > 3$ its absolute value is controlled by the already bounded $(n-3)u^2/r^2$ term. The outer boundary term is nonnegative. This gives the claimed bound on u_r^2, and combining with the tangential part controls $|Du|^2$. Total wave energy is conserved; the estimate concerns energy remaining in a fixed bounded spatial region. □

8.7 Problems and Worked Solutions

These are concise restatements of all 20 source exercises, with editorial solutions. Source problem numbers are retained. Smoothness and boundary assumptions are stated where they affect a conclusion; in Problem 17 connectedness is essential.

猜出乘子仍需验证

Kelvin 变换启发 M，但本稿不据此宣称作用量不变。直接核对 $C_t - \operatorname{div} R = M(u_{tt} - \Delta u)$ 才建立守恒律；负项须用散度项与平方和重写。

1. Oscillatory weak limits. Show $\sin(kx) \rightharpoonup 0$ in $L^2(0,1)$, and that a periodic mixture with proportions $\lambda, 1-\lambda$ and values a, b converges weakly to $\lambda a + (1-\lambda)b$.

Solution. For a C^1 test ϕ, integration by parts bounds $|\int \sin(kx)\phi| \le C_\phi/k$. Uniform L^2 boundedness and density extend convergence to all L^2 tests. For the mixture, its integral over each cell $[j/k, (j+1)/k)$ is its mean divided by k. Subtract the value $\phi(j/k)$ in each cell; uniform continuity bounds the resulting error by $C \sup_{|x-y|\le 1/k} |\phi(x) - \phi(y)| \to 0$. Again use density and uniform boundedness. Neither result gives strong convergence unless the oscillatory variance vanishes. □

2. A drift with a variational weight. Find a Lagrangian for $-\Delta u + D\varphi \cdot Du = f$.

Solution. Take $L(p,z,x) = e^{-\varphi(x)}(\frac{1}{2}|p|^2 - f(x)z)$. Its equation is $-\operatorname{div}(e^{-\varphi}Du) = e^{-\varphi}f$. Division by the positive weight gives the stated equation. The positive exponential factor and its sign are fixed by the drift. □

3. Elliptic regularization of heat flow. Find an energy for $u_t - \Delta_x u - \varepsilon u_{tt} = 0$.

Solution. Use the space–time Lagrangian $L_\varepsilon(p,s,z,x,t) = \frac{1}{2}e^{-t/\varepsilon}(|p|^2 + \varepsilon s^2)$. Its equation is $-\operatorname{div}_x(e^{-t/\varepsilon}Du) - \partial_t(\varepsilon e^{-t/\varepsilon}u_t) = 0$. Canceling the weight gives precisely the asserted equation. This is a variational identity for fixed endpoint variations; a choice of temporal boundary conditions is additional data, not determined by the PDE alone. □

4. Weighted null Lagrangians. Show $L(P,z) = \eta(z)\det P$ is null and its integral depends only on the trace.

Solution. By Piola,

$$\partial_i[\eta(u)(\operatorname{cof} Du)_{\alpha i}] = \eta_{z_\beta}(u)u_i^\beta(\operatorname{cof} Du)_{\alpha i} = \eta_{z_\alpha}(u)\det Du = L_{z_\alpha}.$$

Thus the Euler–Lagrange expression vanishes. The homotopy argument in §8.1 proves boundary dependence. The calculation needs no invertibility or fixed sign of $\det Du$. □

5. The degree is an integer. Let $B = B(x_0, r)$ avoid $u(\partial U)$ and let η be supported in B with integral one. Show $\int_U \eta(u) \det Du \in \mathbb{Z}$.

Solution. Define the distribution $\nu(\psi) = \int_U \psi(u) \det Du$ on B. For $\psi \in C_c^\infty(B)$, the preceding cofactor identity gives

$$\int_U (\partial_\alpha \psi)(u) \det Du = \int_U \partial_i [\psi(u)(\operatorname{cof} Du)_{\alpha i}] = 0,$$

because $\psi(u)$ vanishes near ∂U. Thus all distributional derivatives of ν vanish. A distribution with zero derivatives on a connected ball is constant: mollification makes it a constant smooth function on each smaller ball, and overlapping balls identify the constants. Therefore $\nu(\psi) = d \int_B \psi$.

Use Sard's theorem (an external finite-dimensional theorem) to choose a regular value $y \in B$; a value outside the image is permitted. Its preimages in U form a finite set, since they are isolated by the inverse function theorem and lie in a compact set away from the boundary. For a sufficiently small neighborhood of y, all preimages lie on these finitely many inverse branches; compactness excludes extra branches. Signed change of variables gives $d = \sum_{x \in u^{-1}(y)} \operatorname{sgn} \det Du(x) \in \mathbb{Z}$. Hence $\nu(\eta) = d$. Approximation extends the formula to the stated C^1 density. This proof also shows independence from the chosen normalized η within the same ball. □

度数是带符号计数

正负 Jacobian 的原像分别计 +1 与 −1。先证推前分布在避开边界像的球上为常数，再在正则值附近换元；不能对可能多对一的映射直接套全局一一换元。

6. The curvature integral is a boundary quantity. Show $\int_U (1 + |Du|^2)^{-3/2} \det D^2 u$ depends only on $Du|_{\partial U}$.

Solution. Apply Problem 4 to the map $v = Du : U \subset \mathbb{R}^2 \to \mathbb{R}^2$ and $\eta(z) = (1 + |z|^2)^{-3/2}$. The resulting weighted determinant integral is exactly the displayed expression. The relevant boundary datum is the full gradient of u, not merely u itself. □

7. A quadratic null Lagrangian. Show $L(P) = \operatorname{tr}(P^2) - (\operatorname{tr} P)^2$ is null.

Solution. $L_{P_{\alpha i}} = 2P_{i\alpha} - 2(\operatorname{tr} P)\delta_{\alpha i}$. With $P = Du$, its row divergence is $2\sum_i \partial_i \partial_\alpha u^i - 2\partial_\alpha \sum_i \partial_i u^i = 0$. Notice $\operatorname{tr}(P^2)$ is not the Frobenius norm squared for an arbitrary nonsymmetric matrix. □

8. Why the area functional escapes this direct method. Explain the obstruction for $I[u] = \int \sqrt{1 + |Du|^2}$ in $W^{1,q}$.

Solution. For $q > 1$ there is no coercive lower bound $\alpha|p|^q - \beta$ because the density grows only linearly as $|p| \to \infty$. For $q = 1$, $W^{1,1}$ is not reflexive and an energy bound gives no automatic weakly convergent $W^{1,1}$ subsequence: gradients may concentrate. The functional is convex, so the missing step is compactness/coercivity, not gradient convexity. A BV relaxation and its boundary terms belong to a different theory, not to the $q > 1$ theorem above. □

9. Legendre–Hadamard condition for systems. Derive the rank-one second-variation condition, and exhibit a nonconvex example.

Solution. Use $\phi_\varepsilon = \varepsilon a \rho(x \cdot b/\varepsilon)\zeta(x)$, so $D\phi_\varepsilon = \rho'(a \otimes b)\zeta + O(\varepsilon)$. The scalar triangular-wave proof gives

$$L_{P_{\alpha i} P_{\beta j}}(Du, u, x) a_\alpha a_\beta b_i b_j \geq 0.$$

For 2×2 matrices, $L(P) = \det P$ is affine on every rank-one line, so this expression is identically zero. Yet with $P = \operatorname{diag}(1, -1)$ and $Q = -P$, one has $L(P) = L(Q) = -1$ and $L((P + Q)/2) = 0$, violating convexity. For larger dimensions use a fixed 2×2 minor. □

10. A nonzero semilinear solution by constrained minimization. For $n > 2$ and $1 < q < (n + 2)/(n - 2)$, solve $-\Delta v = |v|^{q-1} v$ with zero trace.

Solution. Minimize $\frac{1}{2}\|Du\|_2^2$ with $\int |u|^{q+1} = 1$. The class is nonempty, and compact $H_0^1 \hookrightarrow L^{q+1}$ preserves its constraint under a minimizing sequence. The first derivative of the constraint evaluated on u equals $q + 1 \neq 0$, so the implicit-curve argument gives $-\Delta u = \lambda |u|^{q-1} u$. Testing by u gives $\lambda = \|Du\|_2^2 > 0$. Then $v = \lambda^{1/(q-1)} u$ is the desired nonzero solution. The constraint is superquadratic, so here its differentiability and compactness use the subcritical embedding rather than the linear-growth hypothesis of §8.4.1. □

11. A monotone nonlinear Robin condition. Assume $0 < a \leq \beta'(s) \leq b$ and $f \in L^2(U)$. Prove weak existence and uniqueness for $-\Delta u = f$, $\partial_\nu u + \beta(u) = 0$.

Solution. Let $B(s) = \int_0^s \beta(t)\,dt$ and minimize on $H^1(U)$

$$E[u] = \tfrac{1}{2}\|Du\|_2^2 + \int_{\partial U} B(Tu)\, dS - \int_U f u.$$

The trace map is continuous to $L^2(\partial U)$ and B has quadratic growth, so the energy is finite. Also $B(s) \geq as^2/4 - C$, and $\|u\|_{H^1}^2 \leq C(\|Du\|_2^2 + \|Tu\|_{L^2(\partial U)}^2)$. The latter follows by a compactness contradiction: a normalized sequence with vanishing gradient tends to a constant on each connected component, and vanishing trace makes every such constant zero. Thus E is coercive. Its convex gradient and boundary terms are weakly lower semicontinuous (a continuous convex functional is weakly lower semicontinuous by separation), giving a minimizer. The weak equation is

$$\int_U Du \cdot D\phi + \int_{\partial U} \beta(Tu)T\phi = \int_U f\phi \quad (\phi \in H^1(U)).$$

For two solutions test their difference; monotonicity gives $\|D(u-v)\|_2^2 + a\|T(u-v)\|_2^2 \leq 0$, hence uniqueness. □

12. Fixed volume gives constant mean curvature. For a smooth area minimizer with fixed trace and $\int_U u = 1$, show its graph has constant mean curvature.

Solution. The integral constraint has nonzero derivative, so the same implicit-curve argument yields $\int Du \cdot D\phi / \sqrt{1+|Du|^2} = \lambda \int \phi$. Thus

$$-\operatorname{div}\left(\frac{Du}{\sqrt{1+|Du|^2}}\right) = \lambda.$$

With $nH = \operatorname{div} N$, where $N = Du/\sqrt{1+|Du|^2}$, $H = -\lambda/n$ is constant. Reversing normal orientation reverses H. □

13. The dual Dirichlet principle. Prove the minimum of $\frac{1}{2}\|Dw\|_2^2 - (f,w)$ over H_0^1 equals $\max_{\operatorname{div}\xi = f} -\frac{1}{2}\|\xi\|_2^2$ over $L^2(U;\mathbb{R}^n)$.

Solution. Distributional divergence and zero trace give $(f,w) = -\int \xi \cdot Dw$. Consequently

$$\tfrac{1}{2}\|Dw\|_2^2 - (f,w) = \tfrac{1}{2}\|Dw + \xi\|_2^2 - \tfrac{1}{2}\|\xi\|_2^2 \geq -\tfrac{1}{2}\|\xi\|_2^2.$$

For the Poisson solution u, take $\xi = -Du$. It is feasible since $\operatorname{div}\xi = f$ and the square vanishes. Thus both extrema are attained and equal; the dual sign is fixed by this choice of ξ. □

对偶通量的正负

原方程 $-\Delta u = f$ 配合对偶约束 $\operatorname{div}\xi = f$，最优通量应取 $\xi = -Du$。完成平方是 $|Dw + \xi|^2$，不是减号。

14. A multivalued obstacle equation. Relate the variational inequality to $-\Delta u + \beta(u - h) \ni f$, where $\beta(s) = 0$ for $s > 0$, $\beta(0) = (-\infty, 0]$, and $\beta(s) = \emptyset$ for $s < 0$.

Solution. Under the local H^2 and continuity regularity described in §8.4.2, let $\mu = -\Delta u - f \geq 0$, supported on $\{u = h\}$. Then $\eta = -\mu \in \beta(u - h)$ and $-\Delta u + \eta = f$ a.e. Conversely these relations give $\int Du \cdot D(w - u) - f(w - u) = \int \mu(w - u) \geq 0$ for feasible w, whenever these function pairings are justified; globally H^2 and $\mu \in L^2$ suffice. For the bare H^1 formulation the precise equivalent is $0 \in -\Delta u - f + N_K(u)$ in H^{-1}, where $\langle \eta, w - u \rangle \leq 0$ defines the dual normal cone. A pointwise multiplier is not asserted for a general distributional reaction. □

15. A gradient constraint. Minimize the obstacle-type quadratic energy on $K = \{u \in H_0^1(U) : |Du| \leq 1 \text{ a.e.}\}$ and derive its inequality.

Solution. This nonempty set is convex and norm closed: strong H^1 convergence has an a.e. convergent gradient subsequence. Hence it is weakly closed. The quadratic energy is coercive and weakly lower semicontinuous, so it has a unique minimizer by strict convexity. Its segment variations give $\int Du \cdot D(w - u) \geq \int f(w - u)$ for all $w \in K$, with the same converse by completing the energy difference. Pointwise gradient bounds are preserved by convex weak closure, not by assuming weak gradients converge pointwise. □

16. A singular harmonic map. For $n \geq 3$ show $u = x/|x| \in H^1(U; \mathbb{R}^n)$ is a weak harmonic map into the sphere.

Solution. Away from zero, $Du = (I - u \otimes u)/r$, $|Du|^2 = (n - 1)/r^2$, and $-\Delta u = (n - 1)u/r^2$. The integral $\int_0^\varepsilon r^{n-3} dr$ is finite exactly for $n > 2$. Deleting a small ball and passing to the limit proves these are the weak first derivatives; its boundary error is $O(\varepsilon^{n-1})$. For the equation, the new spherical normal derivative is zero because u is independent of r. Thus integration by parts has no inner flux, and the integrable right side converges, proving the distributional equation. Uniformly bounded smooth approximations to a bounded H_0^1 test extend the identity to the weak class in (8.12). No energy-minimizing property has been claimed merely from this equation. □

17. Uniqueness of a positive normalized ground state. On connected U, let positive $u, v \in H_0^1$ both minimize $\int |Dw|^2$ with $\int w^2 = 1$. Show $u = v$.

Solution. The Lipschitz composition $w = \sqrt{(u^2 + v^2)/2}$ belongs to H_0^1 and is normalized. In the interior positivity permits $s = u^2/(u^2 + v^2)$ and

$$|Dw|^2 = \frac{u^2 + v^2}{2} \left| s\frac{Du}{u} + (1 - s)\frac{Dv}{v} \right|^2 \leq \tfrac{1}{2}(|Du|^2 + |Dv|^2).$$

Both endpoint energies are minimal, so equality holds a.e. Strict convexity of the square gives $Du/u = Dv/v$ a.e. The minimizers solve the first-eigenvalue equation and are smooth and strictly positive locally, so $D\log(u/v) = 0$ and u/v is constant on connected U. Normalization fixes that constant at one. Without connectedness the conclusion fails: two identical disjoint components allow different positive masses on their ground states. □

正特征函数的单性与连通性

对数比值梯度为零只保证每个连通分支上为常数。不连通区域的多个分支若有相同最低特征值，可重新分配归一化质量，不能推出全域唯一。

18. Ordered conductivities and identical Cauchy data. Assume $0 < a_1 \leq a_2$, $\operatorname{div}(a_i Du_i) = 0$, and on the boundary $u_1 = u_2$, $\partial_\nu u_1 = \partial_\nu u_2$, $a_1 = a_2$. If $Du_2 \neq 0$ a.e., show both coefficients and solutions agree.

Solution. Boundary integration gives $\int a_1|Du_1|^2 = \int a_2|Du_2|^2$. Each u_i minimizes its strictly convex weighted Dirichlet energy with the common trace. Hence

$$\int a_1|Du_1|^2 \leq \int a_1|Du_2|^2 \leq \int a_2|Du_2|^2 = \int a_1|Du_1|^2.$$

Equality gives $\int (a_2 - a_1)|Du_2|^2 = 0$. The integrand is nonnegative and $Du_2 \neq 0$ a.e., so $a_1 = a_2$ a.e., then everywhere by continuity. Uniqueness for the common Dirichlet problem gives $u_1 = u_2$. □

19. Wave momentum. For $u_{tt} - \Delta u + f(u) = 0$, find p^k, j^k with $(p^k)_t - \operatorname{div} j^k = 0$.

Solution. Use the space–time Lagrangian $\mathscr{L} = \frac{1}{2}u_t^2 - \frac{1}{2}|Du|^2 - F(u)$. Spatial translation in direction k gives multiplier u_k. The time component of its Noether current is $p^k = u_t u_k$; the spatial component is $-j^k$, where

$$j^k = u_k Du + \mathscr{L} e_k.$$

Thus Noether's space–time divergence identity gives the desired law. As a direct check, differentiating and substituting the wave equation cancels $u_t u_{tk}$, the spatial Hessian terms and $f(u)u_k$. □

20. Almgren's frequency formula. Let u be harmonic on a neighborhood of $\overline{B_R}$, nonzero on B_R, and $u(0) = 0$. Set $H(r) = \int_{\partial B_r} u^2$, $D(r) = \int_{B_r} |Du|^2$, $a = r^{1-n}H$, $b = r^{2-n}D$, $N = b/a$.

Solution. First $a(r) > 0$: if $u = 0$ on a sphere, the Dirichlet principle makes it zero inside; analyticity of harmonic functions then makes it zero on B_R, a contradiction. Differentiating surface integrals and using $D = \int_{\partial B_r} u u_r$ gives

$$a' = 2r^{1-n} \int_{\partial B_r} u u_r = 2b/r.$$

The $p = 2$ formula (8.16) gives $b' = 2r^{2-n} \int_{\partial B_r} u_r^2$. Cauchy–Schwarz gives $b^2 \le (r/2)ab'$, hence

$$N' = \frac{ab' - ba'}{a^2} = \frac{ab' - 2b^2/r}{a^2} \ge 0.$$

Let $\beta = 2N(R)$ and $\gamma = a(R)/R^\beta$. Then $a'/a = 2N(r)/r \le \beta/r$; integration from r to R yields $a(r) \ge \gamma r^\beta$. For a homogeneous harmonic polynomial of degree ℓ, $N = \ell$ and $a(r)$ is a constant times $r^{2\ell}$, checking the normalization. We explicitly assume harmonicity beyond $\overline{B_R}$ so all endpoint quantities at R are defined; one may instead use any smaller radius. □

8.8 References and Proof Dependencies

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 8, printed pp. 427–494 (local English PDF pp. 444–511). The Chinese scan covers printed pp. 303–349 (PDF pp. 321–367). The exposition and all proofs, solutions, Chinese mathematical annotations and diagrams here are editorial reconstructions. The source theorem numbering is section-local, unlike these notes.

Evans points to Giaquinta and Giaquinta–Hildebrandt for variational calculus, Struwe and Zeidler for variational methods, Ladyzhenskaya–Uraltseva for the lower-semicontinuity proof, and Ball–Murat for fields of extremals. Further pointers include Kinderlehrer–Stampacchia and Friedman for variational inequalities, Hélein and Lin–Wang for harmonic maps, and Rabinowitz for minimax theory. The mountain pass theorem is attributed to Ambrosetti–Rabinowitz (*Journal of Functional Analysis* 14 (1973), 349–381). Noether and wave applications are linked to Reed–Simon, Strauss, Morawetz, Lax–Phillips and Lax; the conformal-square identity is attributed to Costa–Strauss. Exercise 17 carries the source's Belloni–Kawohl attribution, and exercise 18 its Alessandrini attribution. These are reported source pointers, not independently consulted works.

Expanded here. First and second variations; null Lagrangians and Piola cancellation; Brouwer via no retraction; convex lower semicontinuity and direct methods; Sobolev first variations; weak minors and restricted polyconvexity; fields of extremals; interior and boundary H^2 estimates; constrained minimization and multipliers; obstacle inequalities and harmonic maps; pressure recovery conditional on a divergence right inverse; deformation and mountain pass; the subcritical Palais–Smale argument; Noether currents, scaled-energy monotonicity, conformal multiplier and exterior local wave-energy decay. All 20 exercises have editorial solutions.

External inputs and limits. Chapter 5 supplies traces, compactness, extension, difference quotients, Morrey and Sobolev embedding; Chapter 6 supplies elliptic and spectral tools. Weak compactness, Hahn–Banach separation, Riesz representation, Egoroff, dominated convergence, finite-dimensional inverse/implicit function theorems and Banach-space Picard iteration are prerequisites. The De Giorgi–Nash–Moser and Schauder estimates in the smoothness bootstrap are quoted, not proved. The bounded divergence right inverse used for pressure is also an explicit external theorem. Obstacle $C^{1,1}_{\mathrm{loc}}$ regularity is not proved; its pointwise complementarity interpretation is conditional on the stated regularity. Problem 5 invokes Sard's theorem and the elementary fact that distributions with zero derivatives are constant. Problem 20 uses harmonic analyticity from Chapter 2. Our polyconvex discussion uses $q > n$ and the source's restricted $(P, \det P)$ representation; no full elasticity existence theory, self-injectivity, general system smoothness, or critical-exponent Palais–Smale theorem is claimed. Problem 17 explicitly requires connectedness. This chapter has not received independent mathematical review.